

Activity: Filter with AND, OR, and NOT

Scenario

In this scenario, you need to obtain specific information about employees, their machines, and the departments they belong to from the database.

Your team needs data to investigate potential security issues and to update computers.

You are responsible for filtering the required information from the database.

Task 1. Retrieve after hours failed login attempts

Your team is investigating failed login attempts that were made after business hours. You want to retrieve this information from the login activity. You'll identify all unsuccessful attempts after 18:00.

The `login_time` column in the `log_in_attempts` table contains information on when login attempts were made. Office hours end at '18:00'.

The `success` column in the `log_in_attempts` table contains values of `TRUE` or `FALSE` to indicate whether the login was successful. MySQL stores Boolean values as 1 for `TRUE`, and 0 for `FALSE`. This means that `TRUE` is represented as 1, and `FALSE` represented as 0 in the `success` column.

- Use the `AND` operator to retrieve the failed login attempts that occurred after business hours. Replace the `X` and `Y` with the correct values to filter for the records you need:

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_time > '18:00' AND success = 0;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.001 sec)

How many failed login attempts occurred after 18:00?

☒ 19

☐ 44

☐ 39

☒ 20

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FALSE = FAILED = 0

Task 2. Retrieve login attempts on specific dates

Your team is investigating a suspicious event that occurred on '2022-05-09'. You want to retrieve all login attempts that occurred on this day and the day before ('2022-05-08').

The `login_date` column in the `log_in_attempts` table contains information on the dates when login attempts were made.

- Use the OR operator to retrieve the failed login attempts on the specified days. Replace the X and Y with the correct values to filter for the records you need:

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

How many login attempts were made on these two days?

☒ 75

☐ 89

☐ 44

☐ 67

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187	arusso	2022-05-09	00:36:26	MEX	192.168.77.137	0
189	nmason	2022-05-08	05:37:24	CANADA	192.168.168.117	1
190	jsoto	2022-05-09	05:09:21	USA	192.168.25.60	0
191	cjackson	2022-05-08	06:46:07	CANADA	192.168.7.187	0
193	lrodriqu	2022-05-08	07:11:29	US	192.168.125.240	0
197	jsoto	2022-05-08	09:05:09	US	192.168.36.21	0

75 rows in set (0.014 sec)

Task 3. Retrieve login attempts outside of Mexico

Now, your team is investigating logins that did not originate in Mexico, and you need to find this information. Note that the country field includes entries with 'MEX' and 'MEXICO'. You should use the NOT and LIKE operators and the matching pattern 'MEX%'.

- Run the following SQL query to retrieve login attempts that did not originate in Mexico. Replace X with the correct operator and Y with the correct pattern to filter for the information you need:

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1

How many login attempts were made outside of Mexico?

☐ 73

☐ 122

☒ 144

☐ 194

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```
195 | arevitsk | 2022-05-11 | 08:59:13 | CANADA | 192.168.250.78 | 1 |  
196 | acook | 2022-05-10 | 09:56:48 | CAN | 192.168.52.90 | 0 |  
197 | jsoto | 2022-05-08 | 09:05:09 | US | 192.168.36.21 | 0 |  
200 | jclark | 2022-05-12 | 01:11:45 | CANADA | 192.168.91.103 | 1 |  
-----  
144 rows in set (0.001 sec)  
  
MariaDB [organization]>
```

Task 4. Retrieve employees in Marketing

For tasks 4, 5 and 6 you need to retrieve the information from the `department` and `office` columns in the `employees` table.

You can run the following SQL query if you need to view the columns and values in the `employees` table:

Your team is updating employee machines, and you need to obtain the information about employees in the 'Marketing' department who are located in all offices in the East building (such as 'East-170' or 'East-320').

- Write a SQL query to retrieve this information from the `employees` table. Select all columns and include filters on the `department` and `office` columns to return only the needed records.

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

7 rows in set (0.044 sec)

What is the username of the first employee in the Marketing department in the East building?

☒ elarson

☐ fbautist

☐ jclark

☐ alevitsk

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Task 5. Retrieve employees in Finance or Sales

Now, your team needs to perform a different update to the computers of all employees in the Finance or the Sales department, and you need to locate information on these employees.

- Write a SQL query to retrieve records for employees in the 'Finance' or the 'Sales' department.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292

What is the username of the first employee in the Sales department returned by the query?

☒ lrodriqu

☐ bisles

☐ tbarnes

☐ sgilmore

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Task 6. Retrieve all employees not in IT

Your team needs to make one more update. This update was already made to employee computers in the Information Technology department. The team needs information about employees who are not in that department. You should use the `NOT` operator to identify these employees.

- Write a SQL query to retrieve records for employees who are not in the 'Information Technology' department.

```
MariaDB [organization]> SELECT *  
  -> FROM employees  
  -> WHERE NOT department = 'Information Technology';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id   | username | department | office      |  
+-----+-----+-----+-----+-----+  
|          1000 | a320b137c219 | elarson  | Marketing  | East-170    |  
|          1001 | b238c825d303 | hwerone  | Marketing  | Central-276 |
```

How many employees are not in the Information Technology department?

☐ 170

☒ 161

☐ 188

☐ 122

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```
|          1198 | q308r573s459 | jmartine | Marketing  | South-117    |  
|          1199 | r520s571t459 | areyes   | Human Resources | East-100    |  
+-----+-----+-----+-----+-----+  
161 rows in set (0.001 sec)
```